

Project Initialization and Planning Phase

Date	5 july 2026
Team ID	xxxxxx
Project Title	Personlized network assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to develop an AI-powered Personalized Network Assistant that monitors, analyzes, and manages network performance by providing intelligent troubleshooting and personalized recommendations to users.
Scope	The project focuses on creating an intelligent system that uses AI and network monitoring techniques to detect connectivity issues, analyze network behavior, provide real-time assistance, and improve user experience.
Problem Statement	
Description	Current network troubleshooting methods are time-consuming and require technical expertise. Users face difficulties in identifying network problems, optimizing performance, and managing connected devices efficiently.

Impact	Solving these issues will improve network reliability, reduce troubleshooting time, enhance user satisfaction, and provide an efficient automated network management experience.
Proposed Solution	
Approach	Implement an AI-based Personalized Network Assistant that collects network data, analyzes performance parameters, detects issues, and provides customized solutions through a user-friendly interface.
Key Features	🔍 AI-based network issue detection and prediction 📊 Real-time network performance monitoring 💡 Personalized troubleshooting recommendations 🗣️ Voice/text-based interaction with users 🔒 Network security alerts and device management



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processing for AI models and network analysis	Intel/AMD processor or Cloud GPU
Memory	required for running applications and storing temporary data	8 GB RAM
Storage	Storage for user data, logs, and models	512 GB SSD / 1 TB SSD
Software		

Frameworks	AI and backend development frameworks	Python, TensorFlow, Flask/Django
Libraries	AI, data processing, and networking libraries	NumPy, Pandas, Scikit-learn, Socket libraries
Development Environment	Coding and testing environment	Jupyter Notebook, VS Code, PyCharm
Data		
Data	Network traffic data, device information, performance logs	Public network datasets / collected network data (CSV format)